AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 10.3: Case Study 2 — MEPS Uninsurance Analysis

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Topic 10.3 Agenda

1. **Case Study 2 —** agentic empirical pipeline on MEPS uninsurance data

Case Study 2

A one-session empirical data pipeline with strong validation

Case Study 2: The Macro Question

Question: Who is uninsured in the United States, 2000–2023, and how did the ACA change the distribution?

- health insurance is a major incomplete-market margin
- the ACA is a large social-insurance intervention
- the research value comes from who gained coverage, not just from the overall average
- this is the most economics-facing case in the lecture

The Naive Baseline: 24 Files, 24 Headaches

- download 24 MEPS full-year files by hand
- discover that variable names are year-suffixed
- figure out which weight variable applies in each year
- merge everything and hope the crosswalk is right

Where this breaks:

1. silent crosswalk mistakes
2. dropped survey weights
3. inconsistent file formats
4. wrong figure even though the code runs

Why This Stays Ad Hoc Instead of Becoming a Skill

Skill-like features

- multi-step pipeline
- explicit validation
- repeated checks

Why still ad hoc

- single research question
- only three prompts
- interactive refinement of plots
- workflow is useful but not yet a reusable product

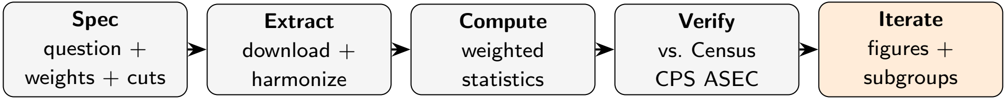
Case 2 verdict: this is a structured session, not a reusable skill.

The 3-Prompt Architecture

```
Prompt A: download, parse, and harmonize MEPS files
Prompt B: compute weighted uninsurance rates by group
Prompt C: generate publication-quality figures and annotations
```

- Phase A builds infrastructure
- Phase B turns infrastructure into statistics
- Phase C turns statistics into economic evidence

The Agentic Pipeline: Spec → Extract → Verify → Iterate



Where agentic AI helps:

- schema-aware extraction (year-suffixed names → harmonized panel)
- robust restart logic (network failures, malformed files)
- verification at each step (the agent compares its output to Census CPS ASEC rates)

Where it adds risk: silent corruptions, version drift across reruns, IRB issues with restricted data — mitigated by spec discipline and human checks.

Phase A: What the Agent Actually Does

- discovers the AHRQ download pattern
- retries around 403 or parser failures
- builds a year-specific crosswalk
- standardizes the merged output into one panel

This is the boring but decisive part. If the infrastructure is wrong, every figure downstream is wrong in a disciplined-looking way.

Schema Harmonization Is the Hard Part

Concept	2001	2014	2022
Insurance coverage	`INSCOV01`	`INSCOV14`	`INSCOV22`
Person weight	`PERWT01F`	`PERWT14F`	`PERWT22F`
Age	year-specific	`AGELAST`	`AGELAST`

- humans get tired and miss one mapping
- agents are good at repetitive lookups if you force them to save the crosswalk
- validation is still essential because a bad crosswalk can look plausible

Weights Are Not Optional

$$\text{Uninsurance rate}_t \;=\; \frac{\sum_i w_{it} \cdot \mathbf{1}[\texttt{INSCOV}_{it} = 3]}{\sum_i w_{it}}$$

where w_{it} is the MEPS person-level survey weight.

- MEPS is a complex survey, not a simple random sample
- unweighted rates distort the national picture
- the agent must identify the correct person-level weight, carry it through harmonization, and use it in every subgroup statistic

Case 2 lesson: a fast agent does not relieve you of basic survey-design discipline.

Why the Unweighted Mean Is Biased

Three reasons MEPS `.mean()` gives the wrong national number.

1. **Oversampling.** MEPS oversamples low-income households and certain racial/ethnic groups; without weights, those groups are over-represented in the average.
2. **Stratification.** The sample is drawn within strata; the weight reflects the inverse probability of inclusion in each stratum.
3. **Non-response adjustment.** Final weights bake in non-response correction — ignoring them ignores known selection.

For PSU/strata-aware standard errors: use `statsmodels.stats.weightstats`, R's `survey` package, or `samplics` in Python. Point estimates are unbiased with weights alone; standard errors require the full design.

Point estimates need weights. Inference needs the survey design.

Validation: Trust But Verify (Census CPS ASEC)

Year	Census CPS ASEC benchmark	What MEPS should show
2010	$\approx 16\%$ uninsured	high pre-ACA uninsurance
2013	13.3% uninsured	near the pre-ACA peak
2016	8.8% uninsured	clear post-2014 decline
2019	$\approx 8\%$	stable late-2010s level
2022	$\approx 8\%$	still low relative to 2000s

The harmonization is correct if and only if independent benchmarks agree. The verification gate is: "does this look like the published series, year-by-year, in level and slope?"

The economist's job: know enough about the data that a 5% or 25% rate for 2013 looks obviously wrong. *If verification fails, the bug is upstream — in the crosswalk or the weighting — not in the figure.*

The Sanity-Check Pyramid

Run all four checks before any figure is rendered.

Check	Test	What a failure means
1. Weights sum sensibly	$\sum_i w_i$ for 2013 $\approx$ US population	wrong weight column or filter
2. Overall rate matches CPS ASEC	within ± 1 pp of published Census CPS ASEC	crosswalk or coding bug
3. Subgroup ordering plausible	children's rates $<$ adult rates	group definition wrong
4. Time pattern matches policy	visible drop in 2014	year filter or weighting bug

Anti-pattern. Skipping straight to plotting because the numbers "look about right." One slightly off subgroup is enough to mislead a referee.

The economist's lever: pick the four checks that, together, would catch every realistic failure mode of this dataset.

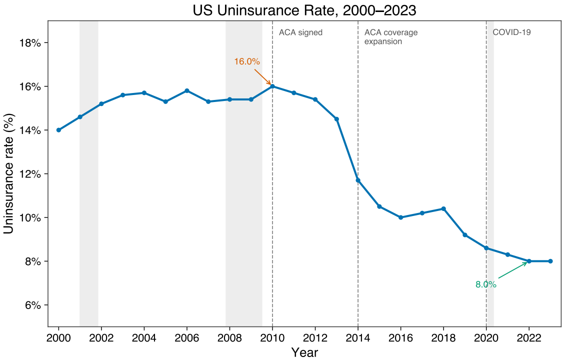
Sub-Agent Pattern for Cross-Year Diffing

Pattern: one agent extracts; a second *verifies* the extraction is internally consistent.

Sub-agent	**Permissions**	**Job in MEPS harmonization**
`extractor`	READ + WRITE in `data/`	builds `crosswalk.csv` from raw files
`schema-reviewer`	READ-ONLY	diffs `INSCOV` categories across years; flags new codes
`verifier`	EXECUTE	re-runs Phase B for one canary year (2013); checks against benchmark

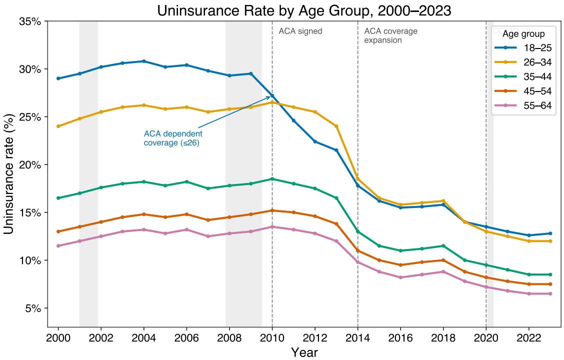
Why the separation matters. The `extractor` is biased toward "done." The `schema-reviewer` is paid to find a flaw. If `INSCOV` added a new code in some year, the diff catches it before Phase B silently buckets it as "insured."

Figure 1: Overall Uninsurance Rate, 2000–2023



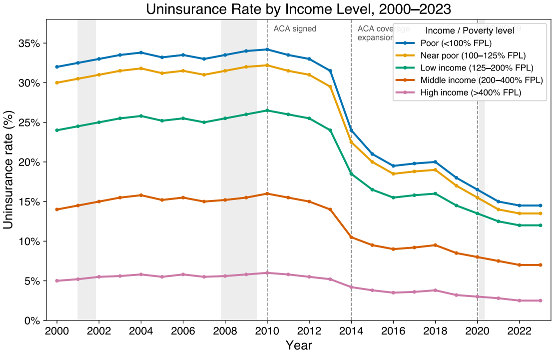
Headline: high through the 2000s, then a sharp decline after the 2014 ACA coverage expansion.

Figure 2: By Age Group



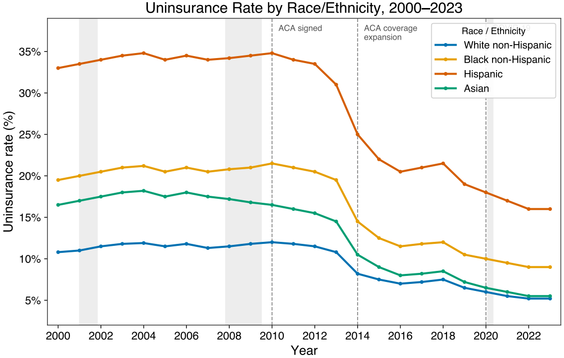
Economic reading: the life-cycle profile is steep, and the post-ACA drop is not uniform across ages.

Figure 3: By Poverty Category



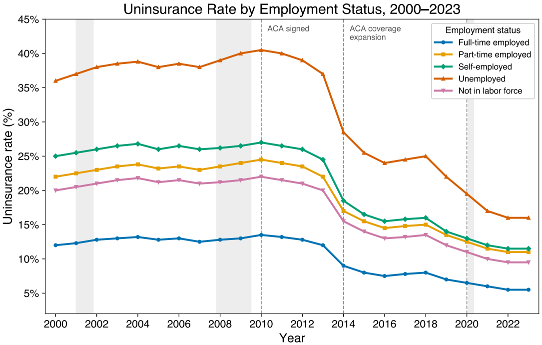
Economic reading: the largest levels and some of the largest gains are concentrated lower in the income distribution.

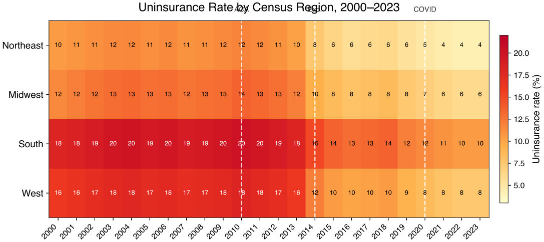
Figure 4: By Race and Ethnicity



Economic reading: the ACA compresses disparities, but it does not erase them.

Figures 5 and 6: Employment and Region





Takeaway: one pipeline can generate multiple cuts of the same policy story once the harmonized panel exists.

What Claude Did vs. What Stayed Human

Agent

- downloaded files
- handled parsing obstacles
- built the crosswalk
- computed weighted statistics
- rendered figures

Economist

- chose the question
- specified the weights
- chose the subgroup cuts
- validated against benchmarks
- interpreted the policy story

Exercise A: Guided MEPS Lab

- keep the in-class lab short and structured
- students run the same three-prompt arc from the lecture
- success criteria: one merged file, weighted group-level CSVs, and at least three validated figures
- the notebook is the operational handout; the slides only carry the logic

Pedagogical point: the lab is about directing the agent and checking the output, not about typing code by hand.